

Bubble Sort

Bubble Sort

- In bubble sort, the idea of the algorithm is to move higher valued elements generally towards the right and lower value elements generally towards the left.

In pseudocode:

- Set swap counter to a non-zero value
- Repeat until the swap counter is 0:
 - Reset swap counter to 0
 - Look at each adjacent pair
 - If two adjacent elements are not in order, swap them and add one to the swap counter

Bubble Sort

Swap Counter
-1



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Bubble Sort

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1	2	3	4	5	6
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- **Worst-case scenario:** The array is in reverse order; we have to “bubble” each of the n elements all the way across the array, and since we can only fully bubble one element into position per pass, we must do this n times.
- **Best-case scenario:** The array is already perfectly sorted, and we make no swaps on the first pass.